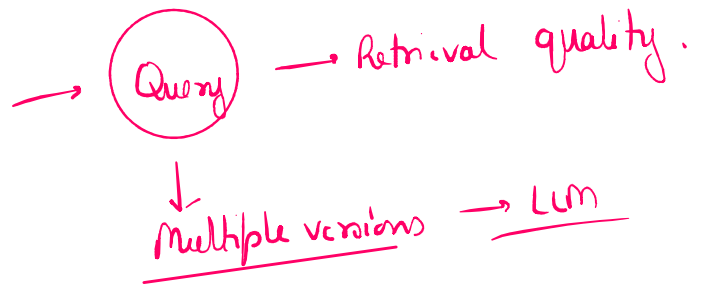


HyDE Hypothetical Document Embeddings

RAG Fusion



2 Phases

① Generate Multiple Queries → Multi Query Retrieval.

② Retrieved results → Fuse / merge → RRF

Top k highest
scoring docs

sort

Reciprocal Rank
Fusion

Query

→ Not well defined

Misses a lot of imp keywords

Simple
Not detailed

Ambiguous

Not Detailed

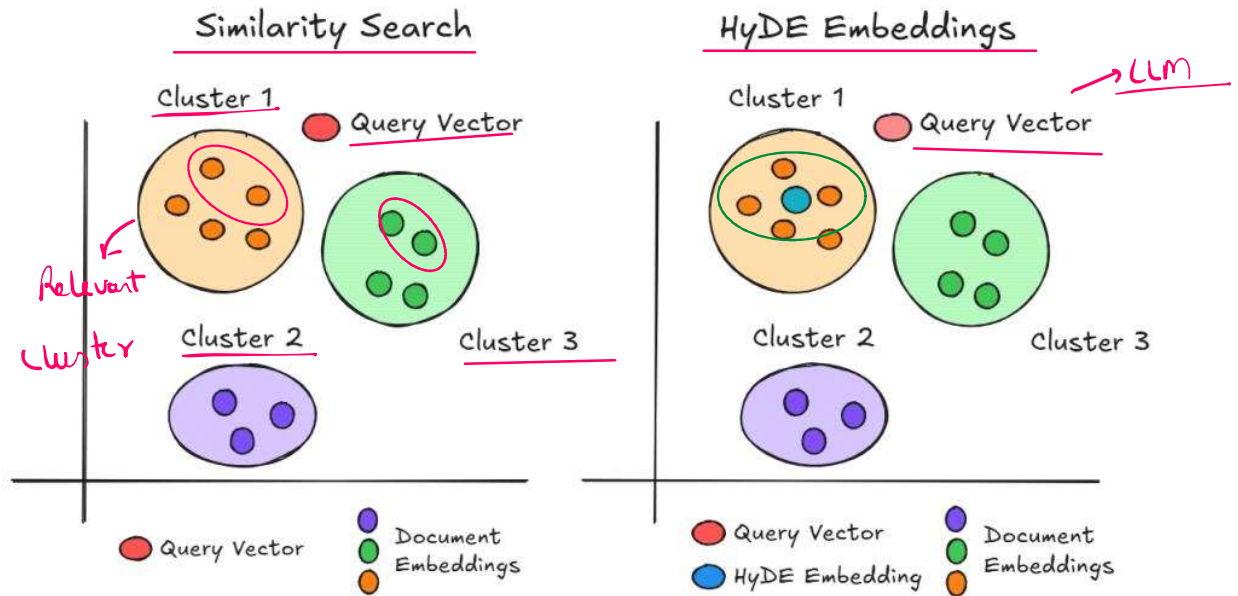
Simple.
detailed X

→ Similarity

↓

Docs - detailed

Apples to oranges



Query $\rightarrow$ Well defined $\times$ Detailed $\times$ $\rightarrow$ (vs) $\rightarrow$ Similarity search

Query $\rightarrow$ semantic meaning $\rightarrow$ Query vector

embedding model.

numerical representation.

Query $\rightarrow$ weak

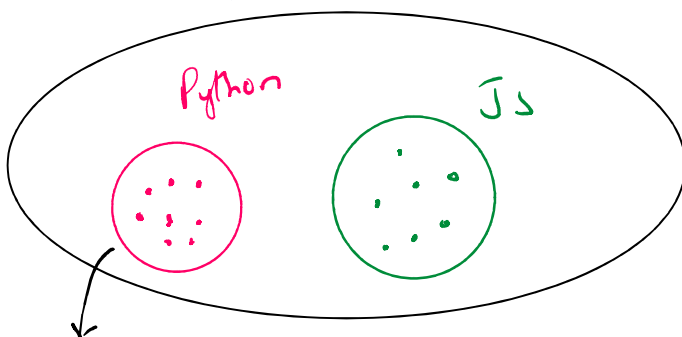
$\downarrow$

$\uparrow$ low quality

Well defined text $\rightarrow$ semantic meaning $\times$

Ill defined text $\rightarrow$ semantic meaning

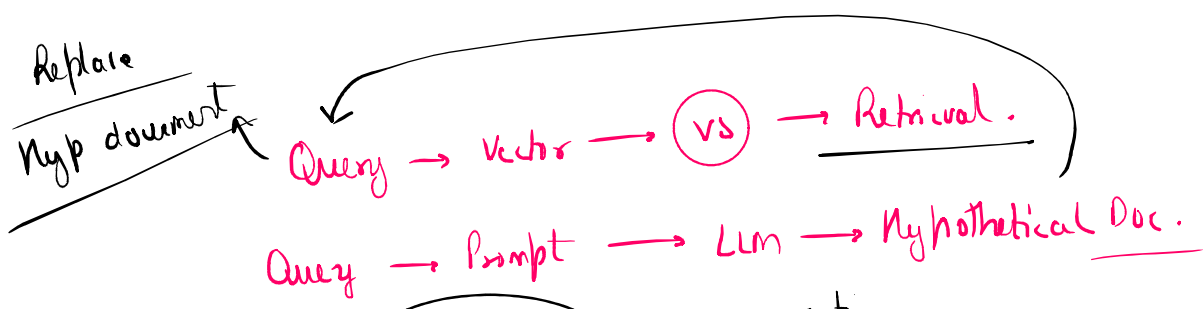
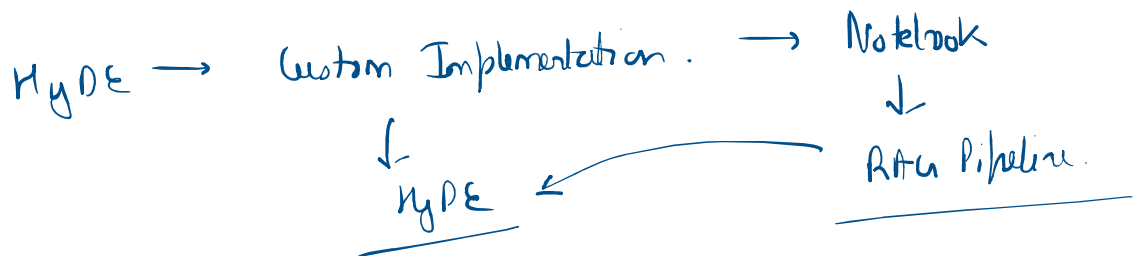
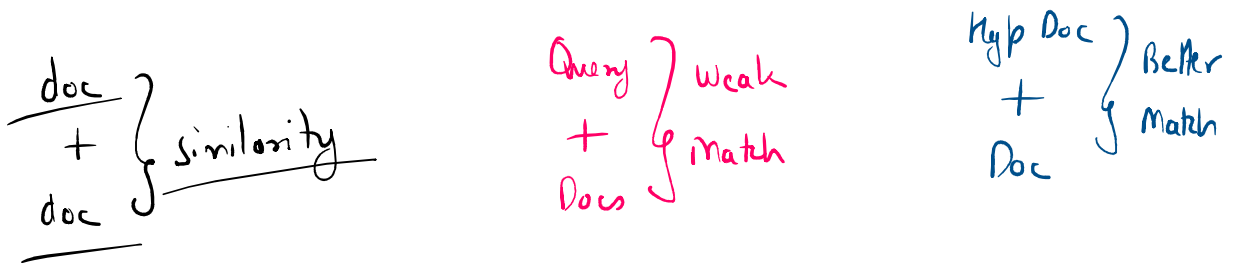
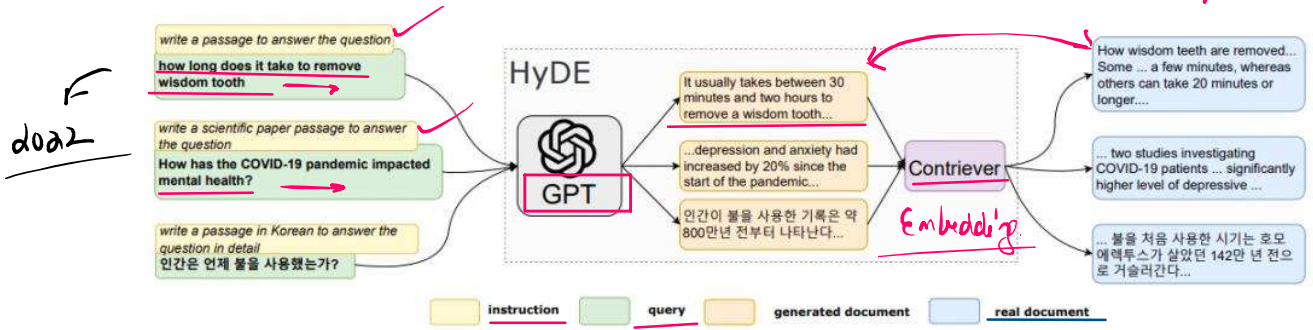
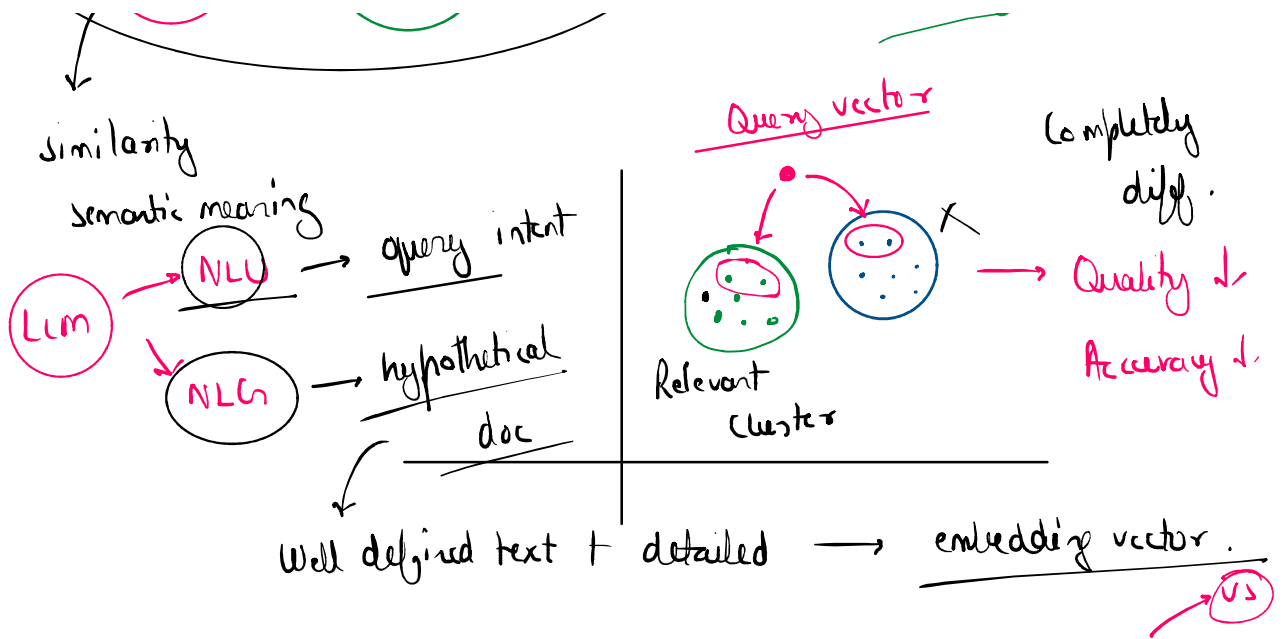
embedding space



Programming lang

Query + Doc $\rightarrow$ Similarity

Query Vector



Query → Prompt → LLM → Hypothetical doc.

